Statistical Significance Testing

Paired bootstrap resampling (1,000 iterations), segment-level chrF++ -- replicating 2025.findings-emnlp.508.pdf Appendix A.10.5

Bhili

Model	ΔchrF++ [95% CI]	p
NLLB	-1.22 [-1.39, -1.06]	<0.001
mT5	+0.30 [0.13, 0.45]	<0.001
Qwen2.5-1.5B	-5.73 [-5.96, -5.51]	<0.001
IndicTrans2	-2.36 [-2.51, -2.20]	<0.001

Table 1: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for Hindi→Bhili.

Model	ΔchrF++ [95% CI]	p
NLLB	-3.84 [-4.01, -3.67]	<0.001
mT5	-0.53 [-0.69, -0.36]	<0.001
Qwen2.5-1.5B	-10.64 [-10.89, -10.40]	<0.001
IndicTrans2	-4.49 [-4.67, -4.31]	<0.001

Table 2: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for Bhili→Hindi.

Model	ΔchrF++ [95% CI]	p
NLLB	-2.59 [-2.75, -2.42]	<0.001
mT5	+0.37 [0.22, 0.52]	<0.001
Qwen2.5-1.5B	-5.04 [-5.19, -4.88]	<0.001
IndicTrans2	-3.41 [-3.56, -3.25]	<0.001

Table 3: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for English→Bhili.

Model	ΔchrF++ [95% CI]	p
NLLB	-2.63 [-2.82, -2.44]	<0.001
mT5	-0.30 [-0.50, -0.13]	0.004
Qwen2.5-1.5B	-5.23 [-5.43, -5.02]	<0.001
IndicTrans2	-2.79 [-2.99, -2.59]	<0.001

Table 4: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for Bhili→English.

Mundari

Model	ΔchrF++ [95% CI]	p
NLLB	-6.61 [-6.82, -6.40]	<0.001
mT5	-0.08 [-0.27, 0.12]	0.432
Qwen2.5-1.5B	-11.08 [-11.32, -10.83]	<0.001
IndicTrans2	-11.46 [-11.70, -11.23]	<0.001

Table 5: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for Hindi→Mundari.

Model	ΔchrF++ [95% CI]	p
NLLB	-7.39 [-7.60, -7.20]	<0.001
mT5	+0.03 [-0.16, 0.23]	0.758
Qwen2.5-1.5B	-22.01 [-22.38, -21.62]	<0.001
IndicTrans2	-10.11 [-10.34, -9.87]	<0.001

Table 6: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for Mundari→Hindi.

Model	ΔchrF++ [95% CI]	p
NLLB	-4.13 [-4.31, -3.97]	<0.001
mT5	-0.25 [-0.40, -0.10]	<0.001
Qwen2.5-1.5B	-8.33 [-8.52, -8.14]	<0.001
IndicTrans2	-7.97 [-8.19, -7.78]	<0.001

Table 7: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for English→Mundari.

Model	ΔchrF++ [95% CI]	p
NLLB	-4.07 [-4.29, -3.86]	<0.001
mT5	-2.32 [-2.53, -2.11]	<0.001
Qwen2.5-1.5B	-6.59 [-6.84, -6.35]	<0.001
IndicTrans2	-4.46 [-4.66, -4.23]	<0.001

Table 8: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for Mundari→English.

Gondi

Model	ΔchrF++ [95% CI]	p
NLLB	-4.34 [-4.64, -4.02]	<0.001
mT5	-0.06 [-0.33, 0.21]	0.670
Qwen2.5-1.5B	-3.63 [-3.95, -3.32]	<0.001
IndicTrans2	-7.68 [-8.00, -7.37]	<0.001

Table 9: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for Hindi→Gondi.

Model	ΔchrF++ [95% CI]	p
NLLB	-8.86 [-9.27, -8.43]	<0.001
mT5	+0.14 [-0.23, 0.52]	0.434
Qwen2.5-1.5B	-11.49 [-11.89, -11.06]	<0.001
IndicTrans2	-11.63 [-12.03, -11.22]	<0.001

Table 10: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for Gondi→Hindi.

Model	ΔchrF++ [95% CI]	p
NLLB	-4.72 [-5.03, -4.43]	<0.001
mT5	+0.02 [-0.25, 0.30]	0.894
Qwen2.5-1.5B	-5.01 [-5.30, -4.71]	<0.001
IndicTrans2	-7.60 [-7.90, -7.31]	<0.001

Table 11: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for English→Gondi.

Model	ΔchrF++ [95% CI]	p
NLLB	-4.78 [-5.15, -4.41]	<0.001
mT5	-1.42 [-1.78, -1.07]	<0.001
Qwen2.5-1.5B	-4.94 [-5.24, -4.59]	<0.001
IndicTrans2	-6.04 [-6.40, -5.70]	<0.001

Table 12: ΔchrF++ (95% CI, p-values) vs. Sarvam-Translate for Gondi→English.